UNITED STATES MARINE CORPS

RTAM, Range Maintenance Branch
Marine Corps Base
PSC Box 20004
Camp Lejeune, North Carolina 28542-0004

SERVICE REQUEST

PART A: Completed by the Government

Number:

CATEGORY **DATE**

ROUTINE URGENT Location of Work:

Description of Work:

PART B: Completed by the Contractor

Date Started:

Description of Work Accomplised:

MATERIAL LIST

ITEM	QTY	COST

MAN HOURS USED

LABOR TYPE	HOURS
Total	0.0

TOTAL 0.00 Date Completed

ATTACHMENT J-1502000-09
RE-OCCURRING RANGE MAINTENANCE LIST

I. Regular Maintenance: **(ALL MINOR CONSTRUCTION NOT TO EXCEED $2000)**

1) Wood Lateral Limit Signs: Fabricate & install new, repair & replace and move existing to new location
2) Aiming stakes (wood & Rumber): Fabricate & install new, repair & replace and move existing to new location
3) Firing Lane Markers: Fabricate & install new, repair & replace and move existing to new location
4) Fill in holes, gouges and depressions on ranges, roads, GPs, TLZs and Training Areas (with earth/fill material)
5) Fabricate, install and repair wood ammo tables
6) Repair & replace metal ammo tables
7) Berm Maintenance (firing line berm or other berm): replace eroded soil, grass etc.
8) Repair wood, concrete and earth firing points
9) Flag Poles: Fabricate & install new, repair & replace and move existing
10) Fill and install sandbags on firing points at ranges
11) Install and remove silt fence
12) Target transportation; move hard targets to and from ranges
13) Fabricate, install and repair flag poles and hardware
14) Change light bulbs
15) De-water holes and craters and fill with earth
16) Repair & replace windsocks and frame
17) Pick up and dispose of garbage, trash or other refuse dumped or abandoned on ranges and training areas
18) Lateral Limit Light Platforms
19) Wood Foot Bridges (Repair/Replace, Fabricate, and install new) NOT TO EXCEED $2000
20) Wood Guard Shacks (Repair/Replace, Fabricate, and install new) NOT TO EXCEED $2000
21) Jersey Barriers (Transportation, Install/Recover, Fill/Empty with water) NOT TO EXCEED $2000
22) Class 3 Property (Temporary Buildings)
23) Fabricate, install, repair, pick up, and dispose of wood, paper, and plastic for targets and/or target facades/bunkers.

ATTACHMENT J-1502000-10
RANGE MAINTENANCE QUARTERLY / ANNUAL INSPECTION CHECKLIST

NAME:	
RANGE/FACILITY	**DATE:**

	Serviceable	Unserviceable	Total #	N/A	Yes	No	Structure #
Range ID Sign w/flag pole							
Flag Pole (flip up)							
Flag Pole (free standing)							
Gate #							
Range Table wood							
Range Table metal							
Lateral Limit Signs:							
Clearly Visable							
Left							
Right							
Center							
Impact Area Sign(s):							
Informational Sign(s):							
Sign(s) Other:							
Targets:							
Clearly Visable							
Infantry #							
Armor #							
Bunker							
Sandbags (protective)							
Wood							
Concrete							
Aiming Stake (wood)							
Aiming Stake (rumber)							
Firing Line Condition:							
Firing Point Earth							
Firing Point Wood							
Firing Point Concrete							
Firing Line Berm							
Protective Berm							
Firing Lane Markers							
Rubber Mulch							
Sandbags (other)							
Facade							
Bullet Trap							
Lateral Limit Light Platforms							
Intracoastal Platform							
Windsock							
SACON Panel Large							
SACON Panel Coping Block							
Fence (wood & wire)							
Fence (chain link or other type)							
Containers							
Vegetation:							
High grass/weeds							
bushes/understory							
Tree							
Vines							

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NAME:	
RANGE/FACILITY	**DATE:**

	Serviceable	Unserviceable	Total #	N/A	Yes	No	Structure #
Bleachers:							
Wood							
Metal							
Cover							
STRUCTURE(S):							
Range Tower							
Classroom							
Head Facilty							
Ammo Break Down							
Shed(s)							
Speaker System							
Bivouac Area							
Parking Area							
ROADS							
Gravel							
Dirt							
Asphalt							
Concrete							
Drainage Ditch							
Culvert							
UTILITIES							
Electric:							
Interior Lights							
Exterior Lights							
Standing Lights							
Recepticles/GFCI							
Switches							
Water:							
Leaks							
Sink							
Toilet							
Well Pump							
Telephone (operational check)							
Telephone Number:							
Erosion							
Rill							
Gully							
Washout							
Silt Fence							
Outlet Stablization							
Vegetation Cover							
Maintenance Issues:							

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ATTACHMENT J-1502000-10-1
TRAINING COURSE INSPECTION CHECKLIST

NAME:	
RANGE/FACILITY	**DATE:**

#	Area	DESCRIPTION	YES	NO	N/A
1A	WOOD TIMBERS	There are no signs of rot, warping, severe weathering, or impact damage.			
1B	WOOD TIMBERS	There are no protruding nails or splinters to cause injury when obstacle is negotiated.			
1C	WOOD TIMBERS	All timbers are securely connected together without excess separation between joints.			
1D	WOOD TIMBERS	All timbers meet specified dimensions as stated in Engineer Drawings or TR 350-6.2			
2A	WALL BOARDS	All boards are securely attached to structure with proper hardware.			
2B	WALL BOARDS	All boards are free of protruding nails, splinters, rot or damage.			
2C	WALL BOARDS	Edges of boards are rounded/smooth where used to support individuals weight.			
3A	HARDWARE	All bolts, nuts, and washers are in place and of the designated type, size and placement.			
3B	HARDWARE	All anchors are made of 3 strand galvanized guy wire or larger.			
3C	HARDWARE	Take up galvanized turnbuckles are used at anchor points of each cable to allow for adjustment.			
3D	HARDWARE	Anchor cables are not used to support obstacles not properly constructed or improperly emplaced in the ground.			
3E	HARDWARE	All cable clamps are positioned with U-bolt placed on the dead or short end of cable.			
4A	FIBER ROPES	All ropes are free of rips, tears, cuts, frays, rot or unraveled sections due to age, excessive wear, or contact with the ground.			
4B	FIBER ROPES	All ropes designed for surmounting are 1.5 inches in diameter.			
4C	FIBER ROPES	Ropes are securely mounted to supporting timbers with ends tied and taped.			
4D	FIBER ROPES	Ends of ropes are tied in a knot or wrapped to prevent fraying.			
4E	FIBER ROPES	Condition/Service logs are maintained on all ropes used for surmounting and suspension.			
5A	DESIGN	Obstacle adheres to blue print specifications.			
6A	FALL PROTECTION	All nets meet ANSI load bearing standard for personnel (ANSI 10.11/OSHA 1926.105) 3.5 inch nylon mesh, 17,500 lb impact resistant.			
6B	FALL PROTECTION	All nets designed for fall protection extend 8 feet out from point of potential fall.			

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#	Area	DESCRIPTION	YES	NO	N/A
6C	FALL PROTECTION	Forged steel hooks are used to fasten net to its supports.			
6D	FALL PROTECTION	Nets are weight tested every 6 months by dropping a 500 lb, 5 cubic feet weight on to it from a height of 25 feet.			
6E	FALL PROTECTION	All nets suspended below high obstacles (in excess of 10 feet) have padding or small mesh material to prevent limbs from penetrating net.			
6D	FALL PROTECTION	Pole vaulting pads are in good conditionwith no tears, holes, or loose material to trip personnel when dismounting.			
6E	FALL PROTECTION	All pole-vaulting pads placed properly at base of designated high obstacles.			
7A	PADDING ON TIMBERS	All padding on timbers are in good condition without signs of damage.			
7B	PADDING ON TIMBERS	Pads are securely attached to the timber supports to prevent movement when impacted.			
8A	BASE CONTAINMENT	Base containment box is adequate to contain all absorbent material located at base of obstacle.			
8B	BASE CONTAINMENT	Containment box does not display signs of rot, damage, instability, or not present.			
8C	BASE CONTAINMENT	Containment box is large enough to dismount obstacle without injury.			
8D	BASE CONTAINMENT	Containment box is filled with either 18” sand, 12 inches of shredded rubber or 24” of sawdust.			
9A	SURFACES	All surfaces beneath low obstacles are free of hazards with the potential to cause harm when crawled upon.			
10A	COURSE CONDITION	Designated course is free of tripping hazards.			
10B	COURSE CONDITION	Course surface is well maintained to prevent injury in case of falls.			
10C	COURSE CONDITION	Course surfaces is raked and policed prior to each use.			
10D	COURSE CONDITION	Course surface is free of large rocks, stones, or concrete materials that cause injury in case of fall.			

DISCREPANCIES:

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ATTACHMENT J-1502000-11
RANGE & TRAINING FACILITY INSPECTION FREQUENCIES

LIVE FIRE RANGES	**INSPECTION FREQUENCY**
A-1	QUARTERLY
B-12	QUARTERLY
D-29A	QUARTERLY
D-29B	QUARTERLY
E-1	ANNUAL
EOD-2	QUARTERLY
EOD-3	QUARTERLY
ETA-1	ANNUAL
ETA-2	ANNUAL
ETA-4	ANNUAL
ETA-5	QUARTERLY
ETA-7	QUARTERLY
ETA-7A	ANNUAL
ETA-7B	ANNUAL
ETA-7C	ANNUAL
ETA-7D	ANNUAL
ETA-8	ANNUAL
ETA-8A	ANNUAL
ETA-9	ANNUAL
ETA-10	ANNUAL
F-11A	QUARTERLY
F-11B	QUARTERLY
F-18	QUARTERLY
F-2	QUARTERLY
F-4	QUARTERLY
F-5	QUARTERLY
F-6	QUARTERLY
G-10 CONVOY OPS SITE 3	QUARTERLY
G-19A	QUARTERLY
G-19B	QUARTERLY
G-21	QUARTERLY
G-27	QUARTERLY
G-27A	QUARTERLY
G-28	QUARTERLY
G-29A	QUARTERLY
G-29B	QUARTERLY
G-29C	QUARTERLY
G-3	QUARTERLY
G-30	QUARTERLY
G-5	QUARTERLY
G-36	QUARTERLY
G-7	QUARTERLY

INDIA-1	QUARTERLY
K-325	QUARTERLY
K-402	QUARTERLY
K-402A	QUARTERLY
K-406A	QUARTERLY
K-406B	QUARTERLY
K-407	QUARTERLY
K-408	QUARTERLY
K-500	QUARTERLY
K-500A	QUARTERLY
K-501	QUARTERLY
K-501A	QUARTERLY
K-502	QUARTERLY
K-503	QUARTERLY
K-503A	QUARTERLY
K-504A	QUARTERLY
K-504B	QUARTERLY
K-505	QUARTERLY
K-506	QUARTERLY
K-507	QUARTERLY
K-508	QUARTERLY
K-509	QUARTERLY
K-510	QUARTERLY
L-5	QUARTERLY
MAC-1	QUARTERLY
MAC-2	QUARTERLY
MAC-3	QUARTERLY
MAC-4	QUARTERLY
MAC-5	QUARTERLY
MAC-6	QUARTERLY
MAC-7	QUARTERLY
R-100	QUARTERLY
GSRA	**INSPECTION FREQUENCY**
SR-6	QUARTERLY
SR-7	QUARTERLY
SR-8	QUARTERLY
SR-9	QUARTERLY
SR-10	QUARTERLY
SR-11	ANNUAL
MOUT	**INSPECTION FREQUENCY**
MOUT UTF (SCR138)	QUARTERLY

MOUT LEJEUNE BLDG #1	QUARTERLY
MOUT SNIPER TOWER	ANNUAL
OP-2 TOWER (OP/SNIPER TOWER)	QUARTERLY
MARSOC	**INSPECTION FREQUENCY**
RR500	ANNUAL
RR501	ANNUAL
RR505	ANNUAL
RR530A	ANNUAL
RR530B	ANNUAL
RR531	ANNUAL
RR532	ANNUAL
SRR534	ANNUAL
SRR534A	ANNUAL
SRR534B	ANNUAL
EOTG	**INSPECTION FREQUENCY**
RR227 SQUARE BAY RANGE	ANNUAL
RR243 3-STORY UTF	ANNUAL
RR249 INDOOR SHOOTING HOUSE	ANNUAL
RIFLE RANGE	**INSPECTION FREQUENCY**
ALPHA	ANNUAL
BRAVO	ANNUAL
CHARLIE	ANNUAL
DODGE CITY	ANNUAL
HATHCOCK	ANNUAL
MECHANICAL	ANNUAL
MULTIPURPOSE	ANNUAL
WALK DOWN	ANNUAL
TRAINING FACILITIES	**INSPECTION FREQUENCY**
AIRFIELD SEIZURE NORTH (CAMP DAVIS NORTH)	QUARTERLY
AIRFIELD SEIZURE SOUTH (CAMP DAVIS SOUTH)	QUARTERLY
CAMP DEVIL DOG MOUT	QUARTERLY
CAMP GEIGER NBC CS CHAMBER	QUARTERLY
CAMP GEIGER MOBILE MOUT	ANNUAL
COMBAT TOWN	QUARTERLY
F-17 RAPPEL TOWER	QUARTERLY
RR211 RAPPEL TOWER	QUARTERLY

RR495 RAPPEL TOWER	QUARTERLY
ASH STREET FOB	QUARTERLY
HAWK FOB	QUARTERLY
MOBILE MOUT FOB	QUARTERLY
GSRA FOB	QUARTERLY
IIT BLDG 904	ANNUAL
LHD DECK	QUARTERLY
MOUT LEJEUNE (EURO CTF)	QUARTERLY
MOUT LEJEUNE BIVOUAC SITE	ANNUAL
MOUT ENHANCEMENT (E-MOUT)	ANNUAL
MOUT MAYOR HOUSE (SCR145)	ANNUAL
MOUT MOBILE	ANNUAL
NBC CS CHAMBER, BLDG 935	QUARTERLY
RR211 EOTG RAPPEL TOWER	ANNUAL
RR495 MARSOC RAPPEL TOWER	ANNUAL
EOTG NON-LETHAL WEAPONS RANGE	ANNUAL
TRAINING SITE ALPHA	QUARTERLY
TRAINING SITE BRAVO	QUARTERLY
ETA-1 BRIDGING	ANNUAL
ETA-1 FIELD FORTIFICATIONS	ANNUAL
ETA-1 OBSTACLES	ANNUAL
MGB BRIDGE DRY GAP SITE (TLZ DOVE)	ANNUAL
F-6 PRACTICE GRENADE RANGE	QUARTERLY
K-510 PRACTICE GRENADE RANGE	QUARTERLY
MILE HAMMOCK BAY DIVE SITE/PIERS	ANNUAL
MOCK UP ROAD RADAR SITE	ANNUAL
M-6 BRIDGE TRAPS BAY ROAD	ANNUAL
INTRACOASTAL WATERWAY SIGNS	ANNUAL
OP-5 MATERIAL STOCKPILE SITE	QUARTERLY
TLZ CONDOR MATERIAL STOCKPILE SITE	QUARTERLY
HOME STATION TRAINING LANES SR-12	ANNUAL

ATTACHMENT J-1502000-11-1
TRAINING COURSE INSPECTION FREQUENCIES

TYPE	OWNING UNIT	LOCATION	STRUCTURE #	INSPECTION	GRID
Obstacle Course	SOI-East	Camp Geiger	SG-945	Quarterly	75001 45451
Obstacle Course	TECOM, MCCSSS	Camp Johnson	M272	Quarterly	78858 44972
Obstacle Course	TECOM, MCCSSS	Camp Johnson	SM290	Quarterly	
Obstacle Course	6th Marines	6th Marines/Softball Field	S154	Quarterly	83640 39227
Obstacle Course	2nd Marines	2nd Marines/Julian C. Smith Rd	SHP232A	Quarterly	84165 38250
Obstacle Course	Marine Corps Engineer School	Courthouse Bay on Ellis Field	SBB62	Quarterly	83563 29700
Obstacle Course	MARSOC	Near RR529	SRR517	Quarterly	75500 30511
Obstacle Course	Weapons Training Bn	Behind RR-11	RR70	Quarterly	76245 29254
Endurance Course	Field Medical Training Bn	Camp Johnson	SM700	Quarterly	78664 46852
Endurance Course	2D MLG	Bldg 858 (Piney Green Rd)		Quarterly	87038 40037
Endurance Course	Marine Combat Training Bn	Camp Devil Dog	SDD12	Quarterly	75088 41245
Endurance Course	2D CEB	2D CEB ETA-9/10		Quarterly	87578 32715
Confidence Course	Marine Combat Training Bn	Camp Devil Dog	SDD12	Quarterly	75088 41245
Confidence Course (NBC)	II MEF CBRN	CS Chamber	N/A	Quarterly	87404 38710
Litter Training Course	Field Medical Training Bn	Camp Johnson	SM99	Quarterly	78533 46723
Leaders Reaction Course (LRC)	G-3/5 RTAM	Vicinity Range F-6	N/A	Quarterly	
Combat Vehicle Operators Course (CVOT)	G-3/5 RTAM	ETA-6/HWY.172	N/A	Quarterly	

Combat Vehicle Operators Course (CVOT)	G-3/5 RTAM	GSRA	N/A	Quarterly	
All Terrain Vehicle Course (ATVC)	G-3/5 RTAM	ETA-3	N/A	Quarterly	
Grenade Assault Course (GAC)	G-3/5 RTAM	F-6	N/A	Quarterly	
Grenade Assault Course (GAC)	G-3/5 RTAM	K-510	N/A	Quarterly	

Serviceability/Unserviceability

Range ID Sign w/flag pole:
The metal sign frame should be free of rust, dents or cracks and be painted black. The metal sign should be of a red (reflective) background and the lettering (various colors) should be easily visible from 50 feet in the daytime and 25 feet during the hours of darkness or decreased visibility. The flagpole shall be easily maneuverable up and down and the locking mechanism shall work properly or as designed. The pulleys shall move or roll freely and without obstruction. The spring-loaded clips should spring back into position after being released and shall be able to clip onto the metal grommets of the flag. The rope shall not be frayed, knotted, or otherwise bunched up so as not to allow the rope movement through the pulleys.

Flag Pole (flip up):
The flip up flagpole shall be easily moved into position by lifting up in a horizontal to perpendicular position using a locking mechanism (usually a metal pin) at the base of the pole to secure it in place. The metal pole should not be dented or crooked when placed into position. The metal surfaces on the flagpole and locking mechanism shall be free of rust and corrosion. Flag clips shall function the same as for the Range ID Sign w/flag pole.

Range Gate (check the following for serviceability): Gate arms are not sagging or listing. Gate arms swing open and close easily or without obstruction.
Gate arms are painted yellow and the paint is not cracking, peeling rusting etc.
Gate is marked with numbers, letters or both; reflective tape present. Tongue on gate arm engages collar or locking enclosure properly. Lock should work properly when tested with the key. Chains are not rusty, broken or missing. Pins (if present) should not be bent or rusty.

Range Table (wood): Serviceable when the table top is not: missing, contain holes, bowing, rotting or unable to bear a 300 Lb. load; Table support members are not missing, rotting or unable to secure the table top. Paint should not be cracking, peeling or faded.

Range Table (metal): Serviceable when the table top is not: missing, bowing, rusting, contain holes or unable to bear a 300 Lb. load; Table support members are not missing, rusting or unable to secure the table top.

Lateral Limit Signs: Lateral limit signs (left, center & right) are unserviceable: when they cannot be seen from the firing line (shall be clearly visible), are lying on the ground, the signs are shot through exposing large holes in the plywood; the paint is faded, crackling or peeling off the wood. The sign post is unserviceable: when the post is broken, cracked, shot through or unable to secure the lateral limit sign.

Signs (Impact Area/informational/other): Signs shall be easily read from a reasonable distance according to the letter size. The sign itself should be free from corrosion, rust or

other substance that would keep the sign from being read and understood. Faded signs shall be replaced when they are no longer readable.

Targets: Targets shall be clearly visible from the firing line without being obstructed by vegetation or earth. Wood, metal and concrete targets are unserviceable when they cannot present a clear picture of itself; have noticeable chips, missing pieces, unable to stand on its own or presents other obvious issues.

Bunker: Concrete or wood bunkers are unserviceable when the roof or walls of the structure no longer provide protection from flying objects destined to damage protected personnel, targets, enclosures, structures etc.

Aiming Stake (wood/Rumber): Aiming stakes are unserviceable when they are unable to stand on their own, listing or falling down, missing or lying on the ground; fragmented or stumps left in the ground or when the Rumber (hard rubber molded to a specific shape or size) covering or the Rumber stake itself presents any of the above symptoms.

Firing Point Earth: The firing point should be free of erosion, excess water, allow for easy access by training units and free of pests.

Firing Point Wood: For outdoor ranges, composition and location vary dependent on the type of range. Wooden blocks, posts, as well as facades should be free of dry rot or damage and the firing point number should be clearly painted and visible. The location of the firing point should be upright and in the correct position allowing personnel to see standing targets, as well as automated targets. Wooden points should be free of protruding nails, wire and should be free of cracks.

Firing Point Concrete: The firing point should be free of cracks, chips or any other form of damage. The firing point number (if required) should be clearly painted and visible. The area around the firing point should be free of erosion, excess water, allow for easy access by training units and free of pests.

Firing line and Protective Berms: The berms should be free of erosion, flooding and covered with grass. Stairs leading to the berm should be in a safe condition for personnel to use, as not to hinder training or be a safety hazard to personnel using the stairs. The stairs should be painted Safety Red and not faded.

Firing Lane Markers: Lane Markers are unserviceable when they cannot be seen from the firing line (shall be clearly visible), are lying on the ground, and if the signs have large holes from being shot through, renders the lane markers illegible; the paint is faded, crackling or peeling off.

Rubber mulch: Rubber mulch shall be replaced with new mulch when the existing rubber mulch is ineffective as a shock absorber. The mulch shall be replaced with the

same material that is currently being used. Mulch spilling out over the containment system shall be shoveled up and disposed of properly.

Sandbags (all): Sandbags are considered unserviceable when the material is broken open, frayed or when soil is spilling out of the sandbag. The sandbags shall be replaced with new cloth or synthetic material. The bag shall be filled approximately three quarters full and secured at the top by a string or cord. The unfilled portion at the top is folded under the main bag and placed on the ground or on top of one another in a line or some other formation that can be used to protect or contain something.

Façade: Wood façades are considered unserviceable when: blown down by wind; they cannot be seen from the firing line properly; missing, pieces laying on the ground, shot through or having large holes in the plywood; paint missing, peeling, cracking or faded. Façade structural members are considered unserviceable when: broken, cracked, shot through, blown down by the wind; cannot secure façade fronts etc.

Bullet Trap: Identify if projectiles can escape the deflection and trap system by observing any holes in the metal plating or trapping system or buckets.

Lateral Limit Light Platforms: The wood structures shall be structurally sound with no missing members, cracked, broken, rotting or decayed wood. The structure shall be able to support one person or 250 Lbs. of dead weight.

Intracoastal Platforms: The wood structures shall be structurally sound with no missing members, cracked, broken, rotting or decayed wood. The structure shall be able to support one person or 250 Lbs. of dead weight.

Windsock: The windsock and frame shall be able to move in 360 degrees without obstruction. The windsock material shall not have any large or small holes in the fabric that will not allow the windsock to turn properly. The unserviceable windsock shall be measured for length, width and opening diameter so that a windsock of the existing size can be procured.

SACON (Shock Absorbing Concrete): SACON panels and coping blocks are unserviceable when it contains areas of wear (surface thinning, cavities and channels) that extend into the structure 50% or more of its original thickness.

Fence (wood and wire): Fences are unserviceable when there is overgrown vegetation, dry rot or damage and has access points between posts (save gates) or allows any access by restricted personnel, or has excessive vegetation or debris on the posts, links or gates. Post should be secure in the ground, upright and plumb to be serviceable. Wiring should be secured tightly between posts with fencing nails and should not have portions missing. Serviceable wooden gates should be secured tightly to the adjacent fence, with appropriate hasps, should have no damaged wood, open and close with no hindrance and properly secure. There should be no holes under the fence or gates as well.

Fence (chain link): Unserviceable when there is excessive rust, holes in the links, allows any access by restricted personnel, or has excessive vegetation or debris on the posts, links or gates. Fence posts should be secure in the ground, upright and plumb to be serviceable. Serviceable chain link gates should be secured tightly to the adjacent gates, with appropriate hasps, should have no damaged links, open and close with no hindrance and properly secure. There should be no holes under the fence or gates as well.

High grass/weeds: Unserviceable when high grass and weeds hinder range operations, range training or traffic (vehicular and foot). This includes, but is not limited to over growth on or covering the view of targets, lateral limit signs, lighting systems, firing points, fences, facades, range gates, range facilities and lane markers. Also unserviceable if the range identification or directional signs cannot be read due to overgrowth and if the grass or weeds exceed one foot on the range proper.

Bushes/understory: Unserviceable when bushes and understory hinder range operations, range training and traffic (vehicular and foot). This includes, but is not limited to over growth on or covering the view of targets, lateral limit signs, lighting systems, firing points, fences, facades, range gates, range facilities and lane markers. Also unserviceable if the range identification or directional signs cannot be read due to overgrowth.

Tree: Unserviceable when trees hinder range operations, range training and traffic (vehicular and foot). This includes, but is not limited to over growth on or blocking the view of targets, lateral limit signs, lighting systems, firing points, fences, facades, range gates, range facilities and lane markers. Also unserviceable if the range identification or directional signs cannot be read due to overgrowth.

Vines: Unserviceable when vines hinder range operations, range training or facility operations. This includes, but is not limited to over growth on range phone boxes, targets, lateral limit signs, lighting systems, firing points, fences, facades, range gates, range facilities and lane markers. Also unserviceable if the range identification or directional signs have excessive overgrowth.

Bleachers (wood): The wood structures shall be structurally sound with no missing members, cracked, broken, rotting or decayed wood seats, treads, supports or base parts. Free of nail pops, missing nails or rusted hangars.

Bleachers (metal): The metal structures shall be structurally sound with no missing members, cracked, broken, fatigued, rusting seats, treads, supports or base parts. The metal structure shall be free of excessive rust, broken, sheared or missing bolts, screws or washers.

Range Towers: Unserviceable if stairs or ladders are not secured, or have damaged steps and rungs. Unserviceable if the tower cannot be secured, has damaged windows, inoperable phone, inoperable light bulbs, as well as fixtures, broken light globes,

damaged GFCI's and covers, overhead cables touching the tower, insect infestation, mold, loose wires or conduit internally and externally. The towers air conditioner/heater should be operable, the cab should be clean, free of leaks and there should be no washouts, ruts or excessive erosion at the base. The light switches should be free of damage, door knobs should work properly, no nail pops, telephone boxes should be secured and free of damage. Gutters and drain spouts shall be free of debris, secured to the tower and shingles should not be missing. All equipment supplementary to the tower should be clean and free of damage.

Classroom: Unserviceable if the classroom cannot be secured, has damaged windows, inoperable phone, inoperable light bulbs, as well as fixtures, broken light bulbs, damaged GFCI's and covers, insect infestation, mold, loose wires or conduit internally and externally. The classrooms air conditioner/heater should be operable, classroom should be clean, free of leaks and there should be no washouts, ruts or excessive erosion at entrances or around the classroom. The light switches should be free of damage, door knobs should work properly, and monitors should be secured and free of damage. Gutters and drain spouts shall be free of debris, secured to the classroom and shingles should not be missing. All equipment supplementary to the classroom should be clean and free of damage.

Head Facility: Unserviceable if the head cannot be secured, has damaged windows, inoperable light bulbs, as well as fixtures, broken light bulbs, damaged GFCI's and covers, insect infestation, mold, loose wires or conduit internally and externally. The head air conditioner/heater should be operable, head should be clean, free of leaks and there should be no washouts, ruts or excessive erosion at entrances or around the head. The light switches should be free of damage and door knobs should work properly. All commodes and urinals should flush properly; the floor drain and sinks should work correctly. Gutters and drain spouts shall be free of debris, secured to the head and shingles should not be missing. All equipment supplementary to the head should be clean and free of damage.

Ammo Breakdown: Unserviceable if the ammo breakdown has inoperable light bulbs, as well as fixtures, broken light bulbs; damaged GFCI's and covers, insect infestation, and loose. The ammo breakdown should be clean, free of roof leaks and there should be no washouts, ruts or excessive erosion around the ammo breakdown. The light switches should be free of damage, gutters and drain spouts shall be free of debris, secured to the ammo breakdown and shingles should not be missing. Ammo breakdown tables should be secured to the base. All equipment supplementary to the ammo breakdown should be clean and free of damage.

Sheds: Unserviceable if the shed cannot be secured, has damaged windows, inoperable light bulbs, as well as fixtures, broken light bulbs, damaged GFCI's and covers, insect

infestation, mold, loose wires or conduit internally and externally. The shed air conditioner/heater should be operable, shed should be clean, free of leaks and there should be no washouts, ruts or excessive erosion at entrances or around the shed. The light switches should be free of damage and door knobs or hasps should work properly. Gutters and drain spouts shall be free of debris, secured to the head and shingles should not be missing. All equipment supplementary to the shed should be clean and free of damage.

Speaker System: The speaker system shall be unserviceable if sound (in the form of voice communication) does not emit from the speakers in a loud or at a high enough decibel rating to be understood by training personnel on the farthest point of the firing line.

Bivouac Area: The bivouac area shall be free from standing water, unusually high grass, overgrown vegetation, trash, rubbish, garbage or other objects that could impede personnel from setting up a tent camp and sleeping inside the established area.

Gravel Roads: Unserviceable if there are washouts, excessive erosion, poor drainage measures, ruts, obstacles blocking thoroughfare, undermined areas, damaged culverts, lack of gravel and any additional gravel road damage that would hinder troop movement, vehicle movement or cause damage to vehicles using the gravel road.

Dirt Roads: Unserviceable if there are washouts, excessive erosion, poor drainage measures, ruts, obstacles blocking thoroughfare, damaged culverts, and any additional dirt road damage that would hinder troop movement, vehicle movement or cause damage to vehicles using the dirt road.

Asphalt Roads: Unserviceable if there are potholes, excessive erosion, poor drainage measures, undermined areas, and obstacles blocking thoroughfare, damaged culverts, and any additional asphalt road damage that would hinder troop movement, vehicle movement or cause damage to vehicles using the asphalt road.

Concrete Roads: Unserviceable if there are holes in the concrete, excessive erosion, poor drainage measures, undermined areas, and obstacles blocking thoroughfare, damaged culverts, and any additional concrete road damage that would hinder troop movement, vehicle movement or cause damage to vehicles using the concrete road.

Drainage Ditch: Unserviceable if there is flooding, excessive erosion or obstructions, as well as excessive debris stopping the water flow of the drainage ditch.

Culvert: Unserviceable if the culvert is damaged, cannot support the flow of water for its location, filled with debris, erosion around the culvert or exposure of the culvert to traffic.

Interior Lights: Unserviceable if light bulbs are inoperable, as well as fixtures, broken bulbs, loose wires or conduit and damaged light switches.

Exterior Lights: Unserviceable if light bulbs are inoperable, as well as fixtures, broken bulbs, loose wires or conduit and damaged light switches.

Standing Lights: Unserviceable if light bulbs are inoperable, as well as fixtures, broken bulbs, loose wires or conduit, improper mounting and damaged light switches.

Receptacles/GFCI: Unserviceable if inoperable, missing covers or damaged.

Switches: Unserviceable if inoperable, missing covers or damaged.

Water: All lines supporting the transfer of water shall be free of leaks. Unserviceable if sinks, commodes, urinals, air handling units, window air conditioning units, overhead water pipes, lift stations and wells exhibit leaks.

Telephone (operational check): The landline telephone (if present on the range) shall be physically (telephone itself) and telephonically (voice communication) checked for damage or communication issues such as the phone cannot call out (no dial tone) or receiving an incoming call by calling the placarded number on the phone. The number on the telephone shall be annotated on the space provided in the inspection checklist.

Erosion (rill): Rill erosion is the removal of soil by concentrated water running through little streamlets causing soil separation.

Erosion (gully): Classical gullies are an advanced stage of channel erosion. They are formed when channel development has progressed to the point where the gully is too wide and too deep to be tilled across. These channels carry large amounts of water after rains and deposit eroded material at the foot of the gully. They disfigure landscape and make land unfit for growing crops.

Washout: The evacuation of earth, gravel or soil by a sudden gush of water as from a rainy downpour. The washout may cause considerable damage to a road and present a safety hazard to personnel, vehicles and equipment.

Silt Fence: Silt fence is used to contain or control sediment from leaving a construction site and infiltrating natural water. The fence is considered unserviceable when the cloth parts are missing, lying on the ground or have big holes in the fabric that allows sediment to pass through the fabric. The stakes supporting the cloth should be perpendicular to the ground and spaced properly in order to not let the cloth parts droop down and let sediment escape by spilling over the fabric and leave the enclosed area.

Outlet Stabilization: A structure designed to control erosion at the outlet of a channel or culvert pipe by reducing the velocity of water flow and dissipating the energy. The structure is usually made of varying sizes of stone placed in asymmetric or geometric

shapes. Missing stones or sediment buildup blocking water flow is considered unserviceable outlet stabilization.

Vegetation Cover: Plants are an excellent source of protection for soil. They not only anchor soil down with their roots, but also intercept rainfall, provide cover from high wind and water velocities with their residues, increase water infiltration, and increase soil aggregation. The most effective erosion controlling plants are perennial grasses and legumes.

Maintenance Inspection Procedure:

The contractor shall use the attached Range Maintenance Inspection Checklists for each Range inspected.

The contractor shall submit the inspection findings to the government in a paper (hardcopy) for each Range inspected. Or, the contractor can submit the inspection findings to the government electronically so that each Range inspected can be printed out individually by the government.

The contractor shall conduct and record the maintenance inspections in enough detail so that the government will be able to create a written Service Request by interpreting the findings and communicate the repair effort needed to effect repairs.

The contractor shall annotate on the checklist if the repair needs to be accomplished as soon as possible (ASAP). This would be an example of an Urgent Service Request.

Inspection Frequency:
The frequency of the Range Maintenance inspections shall be Annually or Quarterly as determined by the Range Inspections Frequency List. Annual inspections to be conducted at the halfway point of the period of performance (POP). Annual inspections to be completed and findings turned into the Government within 14 days of the start date of the annual inspections. Quarterly inspections to be conducted every three months starting with the beginning of the new POP far a total of (4) four inspections per POP. Quarterly inspections to be completed and findings turned into the Government within 30 days of the start date of the Quarterly inspections.

Note:
Discrepancies found during the inspection process shall be annotated with a clear description of the problem so that it can be resolved with corrective maintenance. Discrepancies shall be written on the inspection checklist in the “Maintenance Issues” section of the checklist. The contractor shall provide a hard copy of the completed inspection checklist within one week of the date of the inspection.

RANGE MAINTENANCE INSPECTION CHECKLEST

	Serviceable	Unserviceable	Total #	N/A	Yes	No	Structure #
Range ID Sign w/flag pole							
Range ID Sign							
Range ID Sign Frame							
Flag Pole (flip up)							
Flag Pole (free standing)							
Range Gate ID #_____							
Range Table wood							
Range Table metal							
Lateral Limit Light Platforms							
Intracoastal Platform							
Windsock							
SACON Panel Large							
SACON Panel Coping Block							
SACON Panel Sacrificial Block							
Fence (wood & wire)							
Fence (chain link or other type)							
Containers ISO/Quadcon etc.							
Bullet Trap (stationary)							
Bullet Trap (portable)							
Aiming Stake (wood)							
Aiming Stake (rumber)							
Lateral Limit Signs							
Clearly Visable							
Left							
Right							
Center							
Impact Area Sign(s):							
Informational Sign(s):							
Sign(s) Other:							
Targets							
Clearly Visable							
Infantry #							
Armor #							
Bunker							
Wood							
Concrete							
SACON							
Plastic							
Sandbags							
Other							
Firing Line Condition							
Firing Point Earth							
Firing Point Wood							

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INSPECTION DEFINITIONS & PROCEDURES

	Serviceable	Unserviceable	Total #	N/A	Yes	No	Structure #
Firing Point Concrete							
Firing Line Berm							
Protective Berm							
Micro Terrain Berm							
Firing Lane Markers							
Rubber Mulch							
Sandbags (other)							
Façade							
Wood							
Metal							
Concrete							
SACON							
Plastic							
Sandbags							
Other							
Vegetation							
High grass/weeds							
bushes/understory							
Tree							
Vines							
Bleachers							
Wood							
Metal							
Cover							
Structures							
Range Tower							
Classroom							
Head Facilty							
Ammo Break Down							
Shed(s)							
Speaker System							
Bivouac Area							
Parking Area							
Roads							
Gravel							
Dirt							
Asphalt							
Concrete							
Drainage Ditch							
Culvert							
Utilities							
Electric:							
Interior Lights							
Exterior Lights							
Standing Lights							
Recepticles/GFCI							

	Serviceable	Unserviceable	Total #	N/A	Yes	No	Structure #
Switches							
Water:							
Leaks							
Sink							
Toilet							
Well Pump							
Telephone (operational check)							
Telephone Number:							
Erosion							
Rill							
Gully							
Washout							
Silt Fence							
Outlet Stablization							
Vegetation Cover							

Maintenance Issues:

TRAINING COURSE INSPECTION CHECKLIST

#	Area	DESCRIPTION	YES	NO	N/A
1A	WOOD TIMBERS	There are no signs of rot, warping, severe weathering, or impact damage.			
1B	WOOD TIMBERS	There are no protruding nails or splinters to cause injury when obstacle is negotiated.			
1C	WOOD TIMBERS	All timbers are securely connected together without excess separation between joints.			
1D	WOOD TIMBERS	All timbers meet specified dimensions as stated in Engineer Drawings or TR 350-6.2			
2A	WALL BOARDS	All boards are securely attached to structure with proper hardware.			
2B	WALL BOARDS	All boards are free of protruding nails, splinters, rot or damage.			
2C	WALL BOARDS	Edges of boards are rounded/smooth where used to support individuals weight.			
3A	HARDWARE	All bolts, nuts, and washers are in place and of the designated type, size and placement.			
3B	HARDWARE	All anchors are made of 3 strand galvanized guy wire or larger.			
3C	HARDWARE	Take up galvanized turnbuckles are used at anchor points of each cable to allow for adjustment.			
3D	HARDWARE	Anchor cables are not used to support obstacles not properly constructed or improperly emplaced in the ground.			
3E	HARDWARE	All cable clamps are positioned with U-bolt placed on the dead or short end of cable.			
4A	FIBER ROPES	All ropes are free of rips, tears, cuts, frays, rot or unraveled sections due to age, excessive wear, or contact with the ground.			
4B	FIBER ROPES	All ropes designed for surmounting are 1.5 inches in diameter.			
4C	FIBER ROPES	Ropes are securely mounted to supporting timbers with ends tied and taped.			
4D	FIBER ROPES	Ends of ropes are tied in a knot or wrapped to prevent fraying.			
4E	FIBER ROPES	Condition/Service logs are maintained on all ropes used for surmounting and suspension.			
5A	DESIGN	Obstacle adheres to blue print specifications.			
6A	FALL PROTECTION	All nets meet ANSI load bearing standard for personnel (ANSI 10.11/OSHA 1926.105) 3.5 inch nylon mesh, 17,500 lb impact resistant.			
6B	FALL PROTECTION	All nets designed for fall protection extend 8 feet out from point of potential fall.			

#	Area	DESCRIPTION	YES	NO	N/A
6C	FALL PROTECTION	Forged steel hooks are used to fasten net to its supports.			
6D	FALL PROTECTION	Nets are weight tested every 6 months by dropping a 500 lb, 5 cubic feet weight on to it from a height of 25 feet.			
6E	FALL PROTECTION	All nets suspended below high obstacles (in excess of 10 feet) have padding or small mesh material to prevent limbs from penetrating net.			
6D	FALL PROTECTION	Pole vaulting pads are in good conditionwith no tears, holes, or loose material to trip personnel when dismounting.			
6E	FALL PROTECTION	All pole-vaulting pads placed properly at base of designated high obstacles.			
7A	PADDING ON TIMBERS	All padding on timbers are in good condition without signs of damage.			
7B	PADDING ON TIMBERS	Pads are securely attached to the timber supports to prevent movement when impacted.			
8A	BASE CONTAINMENT	Base containment box is adequate to contain all absorbent material located at base of obstacle.			
8B	BASE CONTAINMENT	Containment box does not display signs of rot, damage, instability, or not present.			
8C	BASE CONTAINMENT	Containment box is large enough to dismount obstacle without injury.			
8D	BASE CONTAINMENT	Containment box is filled with either 18” sand, 12 inches of shredded rubber or 24” of sawdust.			
9A	SURFACES	All surfaces beneath low obstacles are free of hazards with the potential to cause harm when crawled upon.			
10A	COURSE CONDITION	Designated course is free of tripping hazards.			
10B	COURSE CONDITION	Course surface is well maintained to prevent injury in case of falls.			
10C	COURSE CONDITION	Course surfaces is raked and policed prior to each use.			
10D	COURSE CONDITION	Course surface is free of large rocks, stones, or concrete materials that cause injury in case of fall.			

DISCREPANCIES:

ATTACHMENT J-1502000-11-2
INSPECTION DEFINITIONS & PROCEDURES

ATTACHMENT J-1502000-12

Technical Paper 18 **1 September 2016**

DDESB

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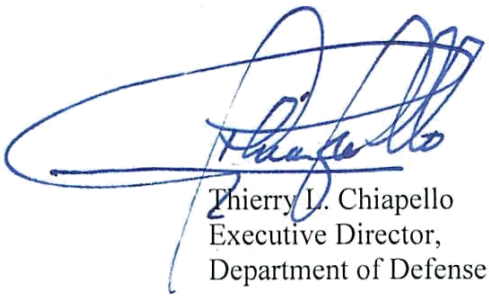
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FOREWORD

Effective 1 September 2016, Technical Paper (TP) 18, "Minimum Qualifications for Personnel Conducting Munitions and Explosives of Concern-Related Activities," supersedes previous versions of TP-18. Existing contracts will continue to apply the provisions of TP-18 dated December 2004 or 1 September 2015 (whichever document was in effect at the initiation of the contract) until completion of the current contract; with exception of options which will use the current effective TP-18 at the time.

Department of Defense Explosives Safety Board (DDESB) TP-18 provides the minimum qualification standards for personnel conducting munitions and explosives of concern (MEC)-related activities in support of the Department of Defense. This TP does not preclude licensing, permitting, training or other defined requirements (e.g., federal, state, local, environmental, etc.).

This document will be updated as necessary.



Thierry L. Chiapello
Executive Director,
Department of Defense Explosives Safety Board

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CHAPTER 1

INTRODUCTION

1.1. GENERAL

a. This technical paper (TP) provides minimum qualification standards, established by the DoD Explosives Safety Board (DDESB), for personnel who:

(1) Support munitions and explosives of concern (MEC)-related activities:

(a) Support workers (SW).

(b) Sweep personnel (SP).

(2) Conduct or supervise MEC-related activities:

(a) Unexploded ordnance technicians (UXOTs).

(b) UXO qualified personnel (UXOQP).

b. MEC-related activities include, but may not be limited to:

(1) Munitions responses to MEC.

(2) Operational range clearance-related activities.

(3) Similar operations that involve the intentional physical contact with MEC including:

(a) Chemical military munitions and chemical agents (CAs) in other than a munitions configuration (collectively referred to as "chemical warfare materiel (CWM)").

(b) Ground-disturbing or other intrusive activities in areas known or suspected to contain MEC.

(4) Performance of escort duties, construction support and anomaly avoidance.

c. During MEC-related activities, personnel may be exposed to explosive or CA hazards (collectively referred to as "explosive hazards") posed by MEC (e.g., UXO, discarded military munitions (DMM)) and associated material potentially presenting an explosive hazard (MPPEH).

d. Personnel must, commensurate with their duties, meet the minimum qualification standards of this TP when:

(1) Performing or supervising operations when intentional physical contact with MEC is planned or will occur.

(2) Conducting ground-disturbing or other intrusive activities in areas known or suspected to contain MEC.

e. Personnel who support, conduct, or supervise MEC-related activities who do not meet the minimum qualification standards of Tables 4.1, 4.2, or 4.3 must not have direct contact with MEC or MPPEH encountered during MEC-related activities until the material has been evaluated and determined not to pose an explosive hazard.

1.2. APPLICABILITY

a. These requirements apply to:

(1) DoD personnel who perform MEC-related activities.

(2) Contractors who perform MEC-related activities under:

(a) DoD contracts.

(b) State and other federal contracts, when the application of these requirements were stipulated or agreed on in property transfer documents (e.g., Finding of Suitability for Early Transfer) and/or agreements (e.g., Environmental Services Cooperative Agreements) between DoD and a State or other federal agency.

(3) DoD personnel or contractors who conduct ground-disturbing or other intrusive activities funded by the DoD in areas known or suspected to contain MEC, and activities involving the intentional physical contact with MEC, except as specified in Paragraph 1.2.b.

b. These requirements do not apply to:

(1) Military explosive ordnance disposal (EOD) personnel or operations.

(2) Personnel performing functions directly related to their assigned duties when conducting research, development, test, and evaluation (RDT&E); munitions management logistics functions (to include: storage, manufacturing, transportation, assembly, testing, inspection, maintenance and demilitarization); munitions operating facility demolition, renovation or maintenance; quality assurance (QA) surveillance testing; and other non-MEC related activities.

(3) Personnel working in support of operational and former ranges, where based on physical or historical evidence, the only munitions-related activities that occurred on the range were ones that involved live-fire training or testing with small arms ammunition. However, such personnel will at a minimum be provided explosives safety education (e.g., DoD recognize,

retreat, report (3Rs) explosives safety education training), including recognition of military munitions.

(4) Other personnel (e.g., commercial developers, contractors) performing MEC-related activities under contract with a State or other federal entities. However, the DDESB recommends the State or federal entity require the application of TP-18 requirements.

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CHAPTER 2

MEC-RELATED ACTIVITIES PERSONNEL – POSITION TITLES, DUTIES AND RESPONSIBILITIES

2.1. GENERAL

a. Training levels and work experience of the following personnel must be documented and verifiable:

(1) MEC-Related Activities Support Personnel

(a) SW

(b) SP (see 29495 in the appendix to this chapter)

(2) Unexploded Ordnance Technicians (UXOT)

(a) UXO-Technician I (UXO-TI) (see 29491 in the appendix to this chapter)

(b) UXO-Technician II (UXO-TII) (by definition are also UXOQP) (see 29492 in the appendix to this chapter)

(c) UXO-Technician III (UXO-TIII) (by definition are also UXOQP) (see 29493 in the appendix to this chapter)

(3) Unexploded Ordnance Qualified Personnel (UXOQP)

(a) UXO Quality Control Specialist (UXOQCS)

(b) UXO Safety Officer (UXOSO)

(c) Senior UXO Supervisor (SUXOS)

(4) Dive-Qualified Personnel

b. DoD activities and DoD contractors performing MEC-related activities are responsible for ensuring personnel employed meet the qualifications specified in this TP.

c. Personnel assigned to support MEC-related activities that are only responsible for supervising (e.g., dive master) or supporting (e.g., dive tender) dive operations from the surface are not required to be a UXOT or meet the criteria of UXOQP. When performing MEC-related activities, such personnel must also meet the qualification criteria for the task performed.

d. Under certain circumstances, the duties of a UXOSO and UXOQCS may be filled by a single individual who meets the training and experience requirements for both positions. If not explicitly specified in the contracting documents or applicable Service guidance, site-specific decisions allowing performance of the duties by a single individual will be provided in writing by the government contracting officer's representative on advice of government explosives safety personnel and in coordination with the contractor.

e. UXOTs and UXOQP must meet the applicable requirements of 18 United States Code (U.S.C.) 842 as amended by the Bureau of Alcohol, Tobacco, Firearms and Explosives in 27 Code of Federal Regulations (CFR) 555.26 and State requirements. Personnel who do not meet these requirements will not be provided access to explosives.

f. Personnel supporting or conducting MEC-related activities may be required to possess specific or specialized licenses (e.g., a commercial or specialized vehicle driver's license, state blasters license), hazardous material endorsements, or be included in a medical monitoring program. (See 29 CFR 1910.120.)

g. Personnel who are working at a hazardous waste site must successfully complete hazardous waste operations and emergency response (HAZWOPER) training per 29 CFR 1910.120.

2.2. DUTIES AND RESPONSIBILITIES. This chapter identifies and describes positions by title and outlines the duties and responsibilities of each position. Although this chapter addresses some qualifications, Chapters 3 and 4 present detailed qualifications and training requirements.

a. MEC-Related Activities Support. Personnel who are not UXOTs or UXOQP, but support MEC-related activities, perform a variety of tasks (e.g., brush-cutting, operating heavy equipment, surveying, site security, dive tenders) required to support the safe conduct of MEC-related activities.

(1) Support Workers (SW)

(a) Must be provided general and site-specific training. At a minimum, this must include:

1. General and site-specific safety. Such training includes, but may not be limited to the proper use of equipment and personal protective equipment (PPE); physical, biological, and chemical hazards; and the potential hazards associated with the tasks they are to perform.

2. Explosives safety training. Recognition of munitions; raw, possibly weathered explosives (e.g., trinitrotoluene (TNT), propellants); and chemical agent identification sets (CAIS) and the actions that should be taken should a suspect munition be encountered (e.g., 3Rs).

(b) SW will:

1. Be protected or escorted when conducting activities (e.g., ground disturbing) that could result in physical contact with MEC, including CWM, CAIS, or MPPEH.

2. Be protected from the potential explosive hazards associated with MEC known or suspected to be present within the site (e.g., shielding heavy equipment operators) when determined necessary by a risk assessment,

3. Be escorted by a UXOQP or a UXOT under the supervision of a UXOQP within areas known or suspected to contain MEC. (Note: Although escort by a UXO-TI is typically performed under the supervision of a UXOQP, the responsible commander or authority may approve UXO-TI personnel to perform escort duties without supervision. Such approval must be based on an approved risk assessment and implementation of methods to mitigate potential exposures). Escorts will help ensure MEC on the surface, and subsurface anomalies are avoided. Support activities performed by SW who may require escort include:

a. Conducting geophysical surveys and similar activities.

b. Clearing vegetation from areas where surface MEC is known or suspected to be present.

c. Operating heavy equipment.

d. Performing site or area security functions requiring access to areas where surface MEC is known or suspected to be present or MEC-related operations are being conducted.

(2) Sweep personnel (SP). The SP must:

(a) Assist UXOT and UXOQP in the performance of MEC-related activities but are not involved in explosive operations.

(b) Be provided general and site-specific training. At a minimum, this must include:

1. General and site-specific safety. Such training includes, but may not be limited to the proper use of equipment and PPE; physical, biological, and chemical hazards; and the potential hazards associated with the tasks they are to perform.

2. Explosives safety training. Recognition of military munitions; raw, potentially weathered explosives (e.g., TNT, propellants); and CAIS and actions that should be taken should a munition or suspect munition be encountered (e.g., 3Rs).

(c) Not be allowed to conduct activities that could result in physical contact with MEC, including CWM or CAIS; material documented as an explosive hazard (MDEH); and

MPPEH, unless the MPPEH has received an initial inspection by UXOQP who determined the material does not pose an explosive hazard.

(d) Be supervised or, if required, escorted by a UXOT (see Paragraph 2.2.b.(1)(k)) or UXOQP within areas known or suspected to contain MEC. The need for an escort is determined by a risk assessment that considers the support or tasks to be performed. Escorts will help ensure MEC and MPPEH on the surface, and subsurface anomalies are avoided.

(e) Be supervised by a UXO-TIII or above when performing activities in areas where there is a medium to high probability that MEC will be encountered, as determined by a risk assessment. Activities that may be performed by SP include:

1. Conducting visual or technology-aided sweeps for surface MEC.

2. Conducting geophysical surveys for subsurface anomalies.

3. Performing, when necessary, field maintenance and/or function checks on geophysical instruments and related equipment within an area known or suspected to contain MEC.

4. Moving MPPEH (e.g., munitions debris and range-related debris) that has received an initial inspection by UXOQP who determined the material does not pose an explosive hazard and is acceptable for further inspection or processing per approved DDESB procedures.

b. UXOT. These personnel perform a variety of MEC-related activities.

(1) UXO-TI. When directed or supervised by a UXOQP, must be able to:

(a) Investigate for and identify MEC and MPPEH, including explosive residues in media (soil, oil, etc.), buildings and installed equipment.

(b) Identify different types of military munitions, including identifying whether a military munition's fuze is armed or unarmed.

(c) Excavate subsurface anomalies for identification.

(d) Move (e.g., consolidate) MEC and MPPEH, within a munitions response site (MRS) or on an operational range after the UXOSO and SUXOS have jointly evaluated and determined the risk of movement to be acceptable.

(e) Operate vehicles transporting explosives, MPPEH or MDEH on site. (Individuals must be appropriately licensed for the class of vehicle being operated.)

(f) Transport military munitions, commercial explosives, and/or MDEH that meets the above criteria that has been determined safe for transport over public traffic routes (PTR).

Such munitions and explosives must be packaged in a manner that allows their safe transport and complies with Department of Transportation (DOT) and other applicable federal and State laws, and DoD policies. (Note: Only UXO determined to be safe for transport by EOD personnel may be transported over a PTR.)

(g) Prepare electric and non-electric firing systems.

(h) Set up decontamination stations and decontaminate CA-contaminated personnel, military munitions and other material of interest (e.g., munitions debris, glass vials) per approved plans. The performance of these functions may require additional training.

(i) Assist UXOQP in documenting the explosives safety status of MPPEH.

(j) Construct engineering controls (protective works).

(k) Escort personnel who are not directly involved in MEC-related activities (e.g., SW, SP, visitors to cultural sites) on property known or suspected to contain MEC, but have an operational requirement and authorization to access such property. Although escort by a UXO-TI is typically performed under the supervision of a UXOQP, the responsible commander or authority may approve UXO-TI personnel to perform escort duties without supervision. Such approval must be based on an approved risk assessment and implementation of methods to mitigate potential exposures.

(2) UXO-TII. Must be able to:

(a) Meet the criteria for and perform the functions of a UXO-TI.

(b) Store explosive materials per applicable guidance, including preparing on-site holding areas to temporarily store and secure MEC or MPPEH, and other explosives (e.g., donor charges).

(c) Determine, using a variety of techniques (e.g., global positioning equipment, land navigation techniques), and record the location of subsurface anomalies, surface MEC and other material of interest in a field environment.

(d) Perform field-collection and -testing procedures to identify explosives-contaminated media or material (e.g., equipment used for the load-assemble-pack of military munitions).

(e) Inspect and document the explosives safety status of MPPEH.

(f) Supervise, as required, SW, SP, and UXO-TI.

(3) UXO-TIII. Must be able to:

(a) Meet the criteria for and perform the functions of a UXO-TI and UXO-TII.

(b) Ensure compliance with a DoD Military Service- and / or DDESB-approved site plans.

(c) Supervise and perform on-site destruction or demilitarization of MEC in place or at a consolidated detonation site. This includes determining where and when it is safe to initiate destruction and when engineering controls are required to mitigate the effects of a detonation.

(d) Implement an explosives storage plan per applicable guidance.

(e) Prepare administrative reports required for munitions responses (e.g., daily UXO team report), operational range clearance activities and similar operations.

(f) Develop and implement standard operating procedures and work plans for munitions responses and operational range clearance activities.

(g) Assist in the preparation of risk and hazard analyses.

(h) Conduct daily site safety briefings.

(i) Supervise the conduct of the different types of MEC-related activities performed at a site.

(j) Determine if MDEH, which is not known or suspected to be UXO, is safe to ship and properly documented for transport over PTR per Technical Bulletin 700-2, Naval Sea Systems Command Instruction 8020.8C, Technical Order 11A-1-47.

(k) Package military munitions, commercial explosives, and MDEH that has been determined safe for transport over PTR.

(l) Serve as UXO team leader.

c. UXOQP. These personnel conduct, manage or oversee MEC-related activities (e.g., reacquire and investigate anomalies, document explosives safety status of materials) required during munitions responses and operational range clearance activities and/or verify the completion of such responses and activities safely and per applicable requirements and approved plans. By definition, UXO-TII and UXO-TIII are considered both UXOTs and UXOQP.

(1) UXOQCS. Must be able to:

(a) Meet the criteria for and perform the functions of a UXO-TIII.

(b) Develop and, upon approval, implement the project's quality control (QC) plan for MEC-related activities per applicable requirements.

(c) Conduct and document QC audits of MEC-related activities for compliance with applicable requirements.

(d) Identify, document, report and ensure completion of corrective actions to ensure MEC-related activities comply with applicable requirements.

(e) Ensure compliance with a DoD Military Service – and/or DDESB-approved site plans.

(f) Prepare QC reports.

(2) UXOSO. Must be able to:

(a) Meet the criteria for and perform the functions of UXO-TIII.

(b) Develop and, upon approval, implement explosives and health and safety plans and programs per applicable DoD, federal, State, and local requirements.

(c) Ensure compliance with a DoD Military Service- and/or DDESB-approved site plans.

(d) Analyze the potential risks (e.g., operational, explosives safety, general safety) associated with MEC-related activities and develop and implement required, mitigating measures.

(e) Establish and ensure compliance with site-specific explosives safety requirements, including, but not limited to:

1. Enforcing personnel limits and explosives safety quantity distance (ESQD) arcs for explosive-related operations.

2. Conducting, documenting, and reporting the results of safety inspections and ensuring implementation of corrective actions.

3. Ensuring protective works and safety equipment within an exclusion zone are used, when required; and operated per manufacturer's specifications, applicable DDESB approvals, DoD policy, and federal, State, or local statutes, regulations, and codes.

(f) Ensure that air-monitoring equipment is operated and maintained properly at sites with known or potential airborne contaminants (e.g., CWM sites).

(g) Evaluate the risk of movement (e.g., consolidation) of MEC within an MRS or on an operational range with the SUXOS, and provide approval for movement by UXOT when the risk of movement is determined to be acceptable.

(3) SUXOS. Must be able to:

(a) Meet the criteria for and perform the functions of a UXO-TIII, UXOQCS, and UXOSO.

(b) Ensure compliance with a DoD Military Service- and/or DDESB-approved site plans.

(c) Plan, coordinate, and supervise all on-site munitions response and operational range clearance activities.

(d) Supervise up to ten UXO teams.

(e) Assist in development of required plans (e.g., health and safety plans).

(f) Review all field reports (e.g., daily reports, audits) and approve UXO team reports.

(g) Evaluate the risk of movement of UXO or DMM within an MRS or operational range with the UXOSO, and provide approval for movement by UXOT when the risk of movement is determined to be acceptable.

d. Dive-Qualified Personnel. Some MEC-related activities require personnel also be dive-qualified. Divers who are performing the duties of a UXOT or UXOQP must meet this TP's criteria for the duties performed and possess the dive-related certifications required for the tasks they are to perform or supervise. An SUXOS, UXOSO, or UXO-TIII providing supervision of MEC-related activities from the surface do not need to possess diver certification.

(1) Dive-qualified personnel will meet the requirements of 29 CFR 1910, Subpart T, and:

(a) Possess dive-related certifications (e.g., dive master, surface-supplied air diver, surface supplied mixed-gas diver), as required for the tasks they are to perform or supervise, from an Association of Commercial Diving Educators (ACDE)-accredited school whose curriculum meets the America National Standards Institute (ANSI) Standard ANSI/ACDE-01;

(b) Have a training certificate with a valid Association of Diving Contractors (ADC) Commercial Diver Certification Card for the appropriate training level; **or**

(c) Completed the underwater portion of NAVSCOLEOD (or foreign equivalent) training.

(2) Divers who are performing MEC-related activities underwater must at a minimum meet the qualifications for a UXO-T1. Such divers must be provided guidance from a SUXOS, UXOSO, or UXO-TIII on the surface. This requires the individual providing the guidance have real-time voice communication with the diver and real-time visual or imaging for confirmation of the material (e.g., military munitions, munitions debris) the diver is encountering. An SUXOS, UXOSO, or UXO-TIII providing supervision of MEC-related activities from the surface do not need to possess diver certification.

APPENDIX

DEPARTMENT OF LABOR, SERVICE CONTRACT ACT DIRECTORY OF OCCUPATIONS - UXO POSITION DESCRIPTIONS[1]

A.1. 29491 UXO-TI

a. Assists in:

(1) Performing reconnaissance and classification of UXO.

(2) Identifying U.S. and foreign guided missiles, bombs and bomb fuzes, projectiles and projectile fuzes, grenades and grenade fuzes, rockets and rocket fuzes, land mines and associated components, pyrotechnic items, military explosives, and demolition materials.

b. Performs location of subsurface UXO using military and/or civilian magnetometers.

(1) Assists in performing excavation procedures on buried UXO.

(2) Performs operator maintenance of military and/or civilian magnetometers.

(3) Locates surface UXO using visual means.

(4) Assists in transporting and storing UXO and demolition materials.

c. Assists in:

(1) Preparing non-electric firing system for a UXO disposal operation.

(2) Preparing electric firing system for a UXO disposal operation disposing of ammunition/ explosives by burning; disposing of ammunition/explosives by detonation.

(3) Operating a personnel decontamination station. Dons and doffs appropriate personal protective equipment in contaminated areas. Assists in the inspection of salvage UXO-related material and erection of UXO-related protective works.

A.2. 29492 UXO-TII

a. Performs reconnaissance and classification of UXO. Identifies U.S. and foreign guided missiles, bombs and bomb fuzes, projectiles and projectile fuzes, grenades and grenades fuzes rockets and rocket fuzes, land mines and associated components, pyrotechnics, military explosives, and demolition materials.

[1] Refer to http://www.dol.gov/whd/regs/compliance/wage/ for the current versions of these descriptions.

b. Locates subsurface UXO using military and/or civilian magnetometers.

(1) Performs excavation procedures on buried UXO by manual means; mechanical means.

(2) Performs operator maintenance of military and/or civilian magnetometers.

(3) Locates surface UXO using visual means. Operates motor vehicle transporting UXO.

(4) Prepares an on-site safe holding area for UXO.

(5) Performs storage of UXO and demolition materials.

(6) Prepares a UXO disposal site.

(7) Prepares non-electric firing system for a UXO disposal operation, electric firing system for a UXO disposal operation, and a detonating cord firing system.

(8) Disposes of UXO/explosives by burning or detonation.

(9) Operates a personnel decontamination station.

(10) Dons and doffs appropriate personal protective equipment in contaminated areas.

c. Inspects salvage UXO-related material.

(1) Erects UXO-related protective works.

(2) Determines a magnetic azimuth using a lensatic compass.

(3) Performs field expedient identification procedures to ID explosive-contaminated soil.

(4) Performs emergency leak seal and packaging of chemical warfare material.

(5) Uses radiographic (x-ray) equipment.

A.3. 29493 UXO-TIII

a. Performs reconnaissance and classification of UXO. Identifies U.S. and foreign guided missiles, bombs and bomb fuzes, projectiles and projectile fuzes, grenades and grenade fuzes, rockets and rocket fuzes, land mines and associated components, pyrotechnic items, military explosives, and demolition materials.

b. Supervises the location of subsurface UXO using military and/or civilian magnetometers.

(1) Supervises the:

(a) Excavation and recovery of subsurface UXO.

(b) Construction of UXO-related protective works.

(c) Location of surface UXO by visual means.

(d) Transporting and storing UXO assuring compliance with federal, State, and local laws.

(e) Disposal of UXO by burning/detonation.

(f) Preparation of a UXO disposal site.

(g) Preparation of an on-site safe holding area for UXO.

(2) Determines UXO-related storage compatibility.

(3) Prepares an explosives storage plan.

(4) Supervises donning and doffing of personal protective equipment, operation of a personnel decontamination station, and maintenance and operator checks on all team equipment.

(5) Prepares UXO-related administrative reports and standard operating procedures.

(6) Conducts daily team safety briefing.

c. Supervises segregation of UXO-related scrap from non-UXO-related scrap, safe-handling procedures, team preventive medicine, and field sanitation procedures.

(1) Performs risk hazard analysis; interpret x-ray of UXO.

(2) Supervises field expedient identification procedures to ID explosives contaminated soil, the determining of a magnetic azimuth using a lensatic compass, emergency leak sealing. and packaging of chemical warfare materiel.

A.4. 29494 UXO SAFETY ESCORT

a. Responsible for the safe escort of non-UXOQP who are not directly involved in specific UXO clearance site work, but have activities to perform within restricted/exclusion areas. Such personnel may include, but are not limited to, contractor personnel involved in the UXO cleanup, cultural visitors, surveying personnel, equipment operators, archaeologists, conservationists, geologists, news media, visiting government personnel, and other personnel as directed.

b. Ensures safety during the transit of persons being escorted by scanning visually in the immediate path of the escorted party, and redirecting the party as necessary to avoid unexploded ordnance and other hazards.

c. Is involved with hazard recognition and avoidance only, not the execution of UXO search or clearance actions.

d. Must have UXO training qualifications.

A.5. 29495 SP

a. Assist UXO personnel in the clearance of UXO, operating only under the direct working supervision of qualified UXO specialist and/or UXO supervisory personnel.

b. Conduct visual and/or instrumented UXO search activities in the field and operate ordnance detection instruments and similar equipment.

c. Remove UXO fuze remnants, fragments and related debris only after such items have been positively identified, inspected and verified as safe to handle by a qualified UXO specialist.

d. Are not involved in the execution of explosives operations.

e. Must have site and job specific contractor training, but does not require UXO qualifications.

CHAPTER 3

UXO-TI TRAINING STANDARDS

3.1. MINIMUM TRAINING STANDARDS FOR UXO-TI

a. This chapter outlines the minimum training standards for a UXO-TI. Personnel who are trained as a UXO-TI must demonstrate the requisite knowledge and ability to perform required tasks in compliance with applicable operational and explosives safety requirements. Candidates for a UXO-TI position must successfully complete:

(1) 200 hours of explosives safety training as described in Paragraph 3.2. or a comprehensive assessment of the individual's ability to successfully perform as a UXO-TI. The course content must convey the information and skills needed to conduct MEC-related activities per applicable requirements. Both the course and comprehensive assessment must include practical exercises evaluating the skills required.

(2) 40 hours of HAZWOPER training per 29 CFR 1910.120.

b. An institution of higher education must provide this training or conduct this assessment and:

(1) Certify successful completion of the course requirements or a comprehensive assessment through written exams and practical exercises.

(2) Have programmatic accreditation by a U.S. Secretary of Education nationally recognized college or university educational accrediting agency, be a component member of a nationally recognized university's system, or be designated by statute as an agency of higher education and have a demonstrated history of providing quality training programs.

c. Before being employed as a UXO-TI, an applicant must provide the perspective employer documentation of successful course completion or successful completion of a comprehensive assessment from an institution of higher education that meets the above criteria. The employer must verify that the individual was trained by the training institution and the institution meets the above requirements.

3.2. CURRICULUM AND TRAINING OBJECTIVES FOR UXO-TI

a. Munitions Responses to MEC

(1) **Description.** Understand the purpose, requirements, and procedures for a munitions response to MEC. This training includes explosives safety and environmental requirements applicable to the conduct of munitions responses to MEC.

(2) **Objectives**

(a) Define terms and identify abbreviations.

(b) Describe the purpose for munitions responses.

(c) Describe planning, safety, and environmental requirements for conducting munitions responses.

(d) Identify the documents (e.g., work plan, accident prevention plan) that should be available for review by field team members.

(e) Describe the typical field tasks of a munitions response and their purpose.

(f) Describe the required process to inspect and document the explosives safety status of MPPEH as either material documented as safe (MDAS) or as MDEH.

(g) Discuss how to and the importance of maintaining a chain of custody for MDAS and MDEH.

(h) Describe the consequences of losing the chain of custody for MDAS and MDEH.

(i) Discuss potential liability concerns associated with evaluating MPPEH and documenting its explosives safety status as MDAS or MDEH.

(j) Describe the proper assembly of protective works.

(k) Describe the general operation of soil sifting, magnetic, and eddy current equipment as a method to separate MEC and munitions debris from soil.

(l) Describe documentation required from a field team conducting MEC activities.

b. Operational Range Clearance Activities and Similar Operations

(1) **Description.** Understand the purpose, requirements, and procedures for the conduct of operational range clearance activities and similar operations. This training includes explosives safety and environmental requirements applicable to the conduct of operational range clearance activities and similar operations.

(2) **Objectives**

(a) Describe the purpose of operational range clearance activities and similar operations and how they differ from munitions responses to MEC.

(b) Describe how operational ranges are designed (e.g., impact areas, safety buffer zones) and used.

(c) Describe planning, safety, and environmental requirements for conducting operational range clearance activities.

(d) Describe the typical field tasks for the conduct of operational range clearance activities and how they differ from munitions responses to MEC.

(e) Describe the potential hazards associated with operational range clearance activities.

(f) Describe documentation required from a field team conducting operational range clearance activities.

(g) Describe the required process to inspect and document the explosives safety status of MPPEH as either MDAS or as MDEH.

(h) Discuss the importance of properly processing and managing MPPEH, MDAS, and MDEH, including the importance of maintaining the chain of custody.

(i) Describe the consequences of losing the chain of custody.

(j) Discuss potential liability concerns associated with evaluating MPPEH and documenting its explosives safety status as MDAS or MDEH.

c. Measurements and Mathematical Computations

(1) Description. Understand the methods for conversion between the U.S. customary and metric systems of measurements and basic mathematical computations.

(2) Objectives

(a) Identify metric prefixes.

(b) Describe mathematical conversions within the metric system.

(c) Convert units from U.S. customary to metric and metric to U.S. customary.

d. Electricity

(1) Description. Understand the basics of electricity and circuitry as applied to MEC-related activities.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Describe electrical conductivity and its characteristics in different materials.

(c) Describe types of cells and batteries, their construction features, and process used to generate electro-motive force.

(d) Describe current flow, factors that affect current flow (including switches) and units of measurement of current flow.

(e) Describe electrical resistance and the factors that affect resistance.

(f) Describe the operation of a series direct current (DC) circuit with respect to Ohm's Law.

(g) Describe the operation of basic parallel DC circuits with respect to the determination of equivalent resistance.

(h) Describe capacitance in terms of charging and discharging a capacitor.

(i) Diagram a basic circuit such as may be used in electric firing system.

e. Physics

(1) Description. Understand the basic physics as applied to MEC-related activities.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Describe forces and how they are graphically represented.

(c) Describe Newton's first and third laws of motion.

(d) Describe the difference between weight and mass.

(e) Describe hydrostatics with respect to fluid pressure.

(f) Describe properties of matter.

(g) Define motion, work, and energy.

(h) Describe measures of and forces affecting motion.

(i) Identify the physical laws affecting gases.

(j) Define magnetism.

f. Military Munitions - Explosives and Explosive Effects

(1) Description. Understand the basics of explosives and explosive effects.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Summarize the history of explosives.

(c) Define propellants, explosives, and pyrotechnics.

(d) Describe characteristics of military munitions (i.e., explosives, propellants, and pyrotechnics).

(e) Identify the types of high explosives groups (primary, secondary, main charge).

(f) Identify forms and classes of propellants and black powder, pyrotechnic, and tracer compositions.

(g) Describe the functioning of an explosive train.

(h) Define explosion and describe types of explosions (detonations, low order detonations, deflagrations).

(i) Describe forms of energy produced by explosions.

(j) Describe effects of an explosion (pressure wave, fragments, debris, thermal, ground shock).

g. Military Munitions - Fuze Functioning

(1) Description. Understand how fuzes function.

(2) Objectives

(a) Define terms and identify abbreviations.

(b) Describe fuze forces.

(c) Describe fundamental principles of fuzes, fuze arming, and firing principles.

(d) Describe fuze components.

(e) Describe methods of employment and uses of fuzes.

(f) Describe typical arming/functioning of fuzes.

(g) Describe fuze types.

h. Explosives Safety Precautions

(1) Description. Understand explosives safety precautions as they apply to the conduct of MEC-related activities, including:

(a) The different categories of MEC (i.e., UXO, DMM, or munitions constituents (MC) that are explosives and in concentration high enough to pose an explosive hazard), and MPPEH and MDEH.

(b) The use of hazards of electromagnetic radiation to ordnance (HERO) safe equipment.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Describe the purpose of explosives safety precautions.

(c) Describe safety considerations that apply by categories of MEC and MPPEH and MDEH.

(d) Describe basic safety precautions for:

1. Explosive-loaded munitions.

2. Submunitions.

3. Toxic chemical-loaded munitions.

4. Pyrotechnic and incendiary munitions.

5. Smoke-loaded munitions.

6. Fuzing systems.

7. Small arms ammunition.

8. Training military munitions.

9. Practice military munitions.

10. Underwater munitions.

i. Military Munitions Identification

(1) Description. Provide a detailed description (e.g., family, group, type) and identify military munition, and applicable specific safety precautions based on type. (See Ammunition Terms.)

(a) Surface-fired, launched, or placed munitions:

1. Projectiles (gun, mortar, howitzer).

2. Small arms ammunition.

3. Land mines and associated components.

4. Infantry rockets and rocket fuzes.

5. Grenade and grenade fuzes.

(b) Air launched or fired munitions:

1. Bombs and bomb fuzes.

2. Guided missiles and missile fuzes.

3. Large rockets and rocket fuzes.

4. Submunitions.

(c) CWM (i.e., chemical munitions and CA in other than munitions configurations).

(d) CAIS.

(e) Smokes and pyrotechnics.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Provide a detailed description and identify military munition, and applicable specific safety precautions based on type.

(c) Demonstrate comprehension and detailed knowledge of live, training, and practice munitions.

(d) Recognize munitions' color codes and markings.

(e) Describe the basic safety precautions for explosive-initiating components.

(f) Describe the safety precautions for munitions by category or group.

j. Underwater Munitions Identification

(1) Description. Understand underwater munitions.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Provide a detailed description and identify military munition, and applicable specific safety precautions based on type.

(c) Demonstrate comprehension and detailed knowledge of live training and practice underwater munitions by category and specific safety precautions based on type.

(d) Recognize munitions' color codes and markings.

(e) Describe the basic safety precautions for explosive-initiating components.

(f) Describe the safety precautions for underwater munitions.

k. Detection Equipment

(1) Description. Understand general, physical, functional, operational, and maintenance of detection equipment for:

(a) Location of subsurface anomalies using various technologies during the conduct of geophysical surveys.

(b) Detection of subsurface anomalies.

(c) Basic geophysical investigation (e.g., anomaly identification, mapping, target discrimination, QC processes).

(2) Objectives

(a) Describe the purpose of geophysical survey equipment; operational characteristics, and capabilities.

(b) Describe the theory of geophysical surveys.

(c) Describe data analysis including the advanced geophysical classification process.

(d) Describe all major and associated components including displays, controls, and indicators.

(e) Describe the purpose of a geophysical system verification and the various tasks involved.

(f) Describe operational tasks and preventive maintenance procedures.

(g) Understand how to inventory and maintain equipment.

(h) Understand typical instrument outputs and their use in munitions responses.

l. PPE

(1) Description. Understand all relevant PPE.

(2) Objectives

(a) Identify where PPE requirements for MEC-related activities are specified and who is responsible for determining whether PPE is required and changing PPE requirements at a site.

(b) Understand the capabilities and limitations of PPE, given the hazards that may be present.

(c) Understand the requirements for employing PPE safely.

(d) Describe the requirements and process for performing decontamination.

m. Demolition Materials

(1) Description. Understand demolition materials and their use including:

(a) Military and commercial explosives (U.S. and foreign).

(b) Initiating components and systems.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Describe military explosives, commercial explosives, and demolition materials and when each might be used.

(c) Describe the purpose of demolition materials and specialized explosive techniques.

(d) Describe tools and equipment used during demolition operations.

(e) Describe demolition accessories.

(f) Describe electric power sources and test sets used with demolition firing circuits.

(g) Describe demolition charge initiators.

(h) Describe demolition charges, charge kits, and assemblies.

(i) Describe safety precautions for preparation and firing of demolition materials.

n. Firing Systems

(1) Description. Understand firing systems and their use.

(2) Objectives

(a) Describe detonating cord demolition procedures.

(b) Describe electric and non-electric firing systems.

(c) Describe safety precautions for preparation and firing of demolition materials, including HERO and static discharge precautions for electric initiators.

(d) Prepare firing systems (both electric and non-electric) for detonation operations.

(e) Describe misfire safety precautions.

(f) Describe hang fire safety precautions.

o. Destruction and Demilitarization

(1) Description. Understand the relationship between destruction (i.e., detonation, burning) and demilitarization requirements, and procedures for the destruction of conventional military munitions.

(2) Objectives

(a) Define terms and identify abbreviations and symbols.

(b) Describe the requirements for and purpose for destruction of conventional munitions.

(c) Describe destruction procedures, including the technologies available for the conduct of contained or controlled destruction.

(d) Describe the authorized destruction methods for different types of military munitions.

(e) Describe requirements and safety precautions for destruction operations.

(f) Describe destruction of conventional explosives and related hazardous materials.

(g) Describe the requirements for demilitarization of military materiel before disposal or recycling.

p. Storage, Handling, and Transportation of Explosives (Military and Commercial)

(1) Description. Understand storage, handling, and transportation of explosives.

(2) Objectives

(a) Describe the purpose for proper storage, handling, and transportation of explosives.

(b) Identify relevant regulations governing storage, handling, and transportation of explosives.

(c) Describe the hazard classification system.

(d) Discuss storage compatibility groups.

(e) Discuss safety requirements.

(f) Discuss ESQD and minimum separation distance requirements.

(g) Discuss transportation requirements for munitions and commercial explosives.

q. Skills Requirements

(1) Description. Demonstrate knowledge of policies, requirements, and procedures in the safe performance of MEC- and MPPEH-related duties.

(2) Objectives. Demonstrate, during a practical exercise, knowledge and comprehension of policies and procedures in safely:

(a) Planning and establishing a standard (100’ x 100’) UXO search grid.

(b) Operating detection equipment used during geophysical surveys within areas known or suspected to contain MEC.

(c) Detecting and determining the coordinates of anomalies.

(d) Reacquiring and investigating (e.g., excavating) an anomaly.

(e) Providing a detailed description of military munitions and identifying specific safety and applicable storage, handling and transportation precautions.

(f) Designing and constructing various types of firing systems (single, dual prime series, dual prime parallel), both electric and non-electric.

(g) Designing, constructing, and detonating a firing system to initiate a complete explosive charge using a line main and/or a ring main.

CHAPTER 4

MINIMUM QUALIFICATION STANDARDS

4.1. Minimum qualification standards for UXOQP, UXOT, and SP and SW are shown in Tables 4.1, 4.2, and 4.3, respectively.

4.2. UXOTs and UXOQP must meet the applicable requirements of 18 United States Code (U.S.C.) 842 as amended by the Bureau of Alcohol, Tobacco, Firearms and Explosives in 27 Code of Federal Regulations (CFR) 555.26 and State requirements. Personnel who do not meet these requirements will not be provided access to explosives.

4.3. Individuals preparing materials for transport or transporting hazardous materials must complete a DOT-compliant (49 CFR 172.704) hazardous materials course.

4.4. Personnel working as UXOTs and UXOQP may have significant breaks between jobs. Only the time personnel have spent working under a contract that is performing MEC-related activities, per Section 1.1, counts towards advancement, with 1880 hours considered a 1-year full-time equivalent. The accumulation of time towards advancement is not limited to time involved in the intentional physical contact with MEC or the conduct of ground-disturbing or other intrusive activities in areas known or suspected to contain MEC, but includes activities in support of a MEC-related contract such as report generation, on–the-job training, equipment maintenance and other activities required to support a contract's requirements.

4.5. A company’s initial assessment of civilian work history or military work experience (e.g.., DD214 and other official records) can be used as the basis for the initial assessment of cumulative MEC related work experience for the logbook.

4.6. UXOTs and UXOQP are responsible for maintaining a logbook of their hours, and should be able to provide reasonable documentation that supports the hours logged. Logbooks should include:

a. Number of work hours

b. Number of MEC-related work hours

c. Type of activity

d. Certification of work hours by authorized supervisor

e. Supervisor and contact information

4.7. UXO-TIs with a 36-month continuous break in the performances of MEC-related activities requires requalification as a UXO-TI based on a comprehensive assessment or supervised on-the-job training. Requalification must be accomplished by attending a UXO-TI requalification

training course or completion of a comprehensive assessment provided by a training organization as outlined in Paragraph 3.1.2.; or by supervised on-the-job training that the employer documents. For other personnel with a 36-month continuous break in the performances of MEC-related activities, when appropriate, provide supervised on-the-job training that is documented by the employer.

4.8. Personnel whose EOD status was terminated for gross negligence in the performance of assigned EOD duties, a flagrant violation of EOD safety procedure or regulation,, or who are not discharged under honorable conditions will not be considered EOD qualified and will not be considered a military EOD School graduate. To obtain employment, such personnel must complete the training required for a UXO-TI.

Table 4.1. Minimum Qualifications for UXOQP[1, 2, 3, 4]

Position Description	Training Required	Minimum MEC-related Experience	Minimum MEC-Supervisory Experience	Minimum Total EOD/MEC Experience
SUXOS	Notes 5, 7, 8, 9, 10, and 11	2 years	1 year	10 years
	Notes 6, 7, 8, 9, 10, and 11	10 years	5 years	13 years
UXOSO	Notes 5, 7, 8, and 9	1 year	0.5 year	8 years
	Notes 6, 7, 8, and 9	8 years	2 years	10 years
UXOQCS	Notes 5, 7, 8, 10, and 11	1 year	0.5 year	8 years
	Notes 6, 7, 8, 10, and 11	8 years	2 years	10 years
Dive Qualified	Note 12	As indicated above for the position description.		

Notes:

1. By definition, UXO-TII and UXO-TIII are UXOQP (minimum qualifications for UXO-T are provided on Table 4.2.
2. Graduate of an Occupational Safety and Health Administration (OSHA)-compliant (29 CFR 1910.120) 40-hour HAZWOPER course.
3. Limited to performance of MEC-related activities as a UXOT or UXOQP or similar civilian government service (e.g., Ordnance and Explosives Safety Specialist (OESS)). Conduct of activities performed as an SP or SW are not considered MEC-related experience and are not counted toward the experiential requirements for UXOQP.
4. Limited to experience in UXO supervisory positions (i.e., UXO-TIII, UXOQCS, UXOSO, OESS).
5. Graduate of a military EOD School of the United States, Canada, Great Britain, Germany, or Australia. (See Paragraph 4.8) for EOD personnel who were terminated for gross negligence in the performance of assigned duties, a flagrant violation of EOD safety procedure or regulation or who are not discharged under honorable conditions will not be considered EOD qualified and will not be considered a military EOD School graduate. To obtain employment, such personnel must complete the training required for a UXO-TI.
6. Graduate of a UXO-TI Course (see Chapter 3), the EOD assistant's course or pass a comprehensive assessment.
7. Graduate of an OSHA-compliant (29 CFR 1910.120(e)(4)) 8-hour Management and Supervisor Training course, if supervising other personnel.
8. Possesses an understanding of applicable explosives safety criteria and experience in the various phases of a munitions response to MEC or the conduct of range clearance activities, as appropriate for the operations to be performed.
9. Must have completed a 10-hour OSHA Construction Safety and Health Training and earned a Department of Labor Construction Safety Course Completion Card.
10. UXOQCS must have either:
    a. Successfully completed training as a quality professional (i.e., International Standards Organization 9001 internal auditor, American Society of Quality Certified quality auditor);
    b. Possess a quality-professional certification by a recognized organization (e.g., U.S. Army the Corps of Engineers and Naval Facility Engineering Command Training Course Construction Quality Management for Contractors; or
    c. Receive company- and project-specific QC training and work under the supervision of a certified quality professional.
11. UXOQCS must demonstrate an understanding of QC and QA practices associated with MEC-related activities and managing and processing MPPEH, including documentation of its explosives safety status.
12 Divers who are independently performing the duties of a UXOT or UXOQP must:
    a. Meet this TP's criteria for the duties performed and the requirements of 29 CFR 1910, Subpart T.
    b. Possess the dive-related certifications required for the tasks they are to perform or supervise. (Note: Certifications (dive or training) must be from an accredited school and meet the requirements contained in ANSI/ACDE Standard-01; be documented as valid by an ADC Commercial Diver Certification Card for the appropriate training level; and/or have documentation of successful completion of an appropriate level of training from an ACDE accredited school or have completed the underwater portion of NAVSCOLEOD (or foreign equivalent) training.)

Table 4.2. Minimum Qualifications for UXOT[1, 2, 3]

Position Description	Training Required	Minimum MEC-related Experience	Minimum Total EOD/MEC Experience
UXO-TIII	Notes 4, 6, and 7	1 years	8 years
	Notes 5, 6, and 7	8 years	8 years
UXO-TII	Notes 4 and 7	0 years	1.5years
	Notes 5 and 7	3 years	3 years
UXO-TI	Notes 4 and 7	0 years	0 years
	Notes 5 and 7	0 years	0 years
Dive Qualified	Note 8	As indicated above for the position description.	

Notes:

1. By definition, UXO-TII and UXO-TIII are also UXOQP.
2. Graduate of an OSHA-compliant (29 CFR 1910.120) 40-hour HAZWOPER course.
3. Limited to performance of MEC-related activities as a UXOT or UXOQP or similar civilian government service (e.g., OESS). Conduct of activities performed as an SP or SW are not considered MEC-related experience and are not counted toward the experiential requirements for UXOQP.
4. Graduate of a military EOD School of the United States, Canada, Great Britain, Germany, or Australia. (See Paragraph 4.8. for EOD personnel who were terminated for gross negligence in the performance of assigned duties, a flagrant violation of EOD safety procedure or regulation or who are not discharged under honorable conditions will not be considered EOD qualified and will not be considered a military EOD School graduate. To obtain employment, such personnel must complete the training required for a UXO-TI.
5. Graduate of a UXO-TI Course (see Chapter 3), the EOD assistant's course or pass a comprehensive assessment.
6. Graduate of an OSHA-compliant (29 CFR 1910.120(e)(4)) 8-hour Management and Supervisor Training course.
7. On-the-job training including, but not limited to familiarity with the process, procedures, and equipment (e.g., geophysical) used for conducting MEC-related activities.
8. Divers who are independently performing the duties of a UXOT or UXOQP must:
   a. Meet this TP's criteria for the duties performed and the requirements of 29 CFR 1910, Subpart T.
   b. Possess the dive-related certifications required for the tasks they are to perform or supervise. (Note: Certifications (dive or training) must be from an accredited school and meet the requirements contained in ANSI/ACDE-01; be documented as valid by an ADC Commercial Diver Certification Card for the appropriate training level; and/or have documentation of successful completion of an appropriate level of training from an ACDE accredited school, or have completed the underwater portion of NAVSCOLEOD (or foreign equivalent) training.)

Table 4.3. Minimum Qualifications for SP and SW[1]

Position Description	Training Required	Minimum MEC-related Experience[3, 4]
SP	Note 2	0 years
SW	Note 2	0 years

Notes:
1. Job- and site-specific training including, but not limited to, general and site specific safety (e.g., proper use of equipment and PPE, physical, biological, and chemical hazards); explosives safety training (e.g., recognition of military munitions, 3Rs
2. Personnel who are working at a hazardous waste site require successful completion of an OSHA-compliant (29 CFR 1910.120) HAZWOPER course.
3. Experience as an SP or SW is not required for UXO-TI certification.
4. Activities performed as an SP or SW are not counted toward the experiential requirements for a UXOT or UXOQP.

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GLOSSARY

G.1. ACRONYMS

3Rs	recognize, retreat, report
ACDE	Association of Commercial Diving Educators
ADC	Association of Diving Contractors
ANSI	America National Standards Institute
CA	chemical agent
CAIS	chemical agent identification sets
CFR	Code of Federal Regulations
CWM	chemical warfare materiel
DC	direct current
DDESB	Department of Defense Explosives Safety Board
DMM	discarded military munitions
DOT	Department of Transportation
EOD	explosive ordnance disposal
ESQD	explosive safety quantity distance
HAZWOPER	hazardous waste operations and emergency response
HERO	hazards of electromagnetic radiation to ordnance
MC	munitions constituents
MDAS	material documented as safe
MDEH	material documented as an explosive hazard
MEC	munitions and explosives of concern
MPPEH	material potentially presenting an explosive hazard
MRS	munitions response site
OESS	Ordnance and Explosives Safety Specialist
OSHA	Occupational Safety and Health Administration
PPE	personal protective equipment
PTR	public traffic route
QA	quality assurance
QC	quality control
RDT&E	research, development, test and evaluation
RSP	render-safe procedures

SP	sweep personnel
SUXOS	Senior UXO Supervisor
SW	support worker
TNT	trinitrotoluene
TP	technical paper
U.S.C.	United States Code
UXO	unexploded ordnance
UXOQCS	UXO Quality Control Specialist
UXOQP	UXO Qualified Personnel
UXOSO	UXO Safety Officer
UXOT	UXO Technician
UXO-TI	UXO Technician I
UXO-TII	UXO Technician II
UXO-TIII	UXO Technician III

G.2. DEFINITIONS

Ammunition Terms. The below provide general meanings for the terms listed. Based on use, certain terms may have a different meanings.

- Family - weapon system munitions which provide a similar capability (e.g., tank, artillery, bombs)
- Group - family broken into more specific category usually by weapon system (e.g., M1 Abrams 120mm, 105mm)
- Type - denotes payload or capability delivery (e.g., high explosive, armor piercing)
- Category:

    - Operational, combat, training, or testing
    - Storage category
    - MEC category (i.e., UXO, DMM or MC (explosive))

- Groups - sensitivity of storage compatibility, munitions with similar explosive characteristics relating to means of detonation, blast considerations, and special containment requirements (e.g., sensitivity to heat, friction, percussion)

DMM. Defined in 10 U.S.C. 2710(e)(2).

explosive. Defined in DoD 6055.09-M, Volume 8.

explosive hazard. Defined in DoD 6055.09-M, Volume 8.

EOD. Defined in DoD 6055.09-M, Volume 8.

EOD personnel. Uniformed military personnel who have graduated from the Naval School, Explosive Ordnance Disposal; are assigned to a military unit with a Service-defined EOD mission; and meet Service and assigned unit requirements to perform EOD duties. EOD personnel have received specialized training to address explosive and certain CA hazards during both peacetime and wartime. EOD personnel are trained and equipped to perform render safe procedures (RSP) on nuclear, biological, chemical, and conventional munitions, and on improvised explosive devices.

explosives or munitions emergency response. Defined in 40 CFR 260.10.

MC. Defined in 10 U.S.C. 2710 (e)(3).

MDAS. Defined in DoD Instruction 4140.62.

MDEH. Defined in DoD Instruction 4140.62.

MEC. Specific categories of military munitions that may pose unique explosives safety risks when present in high enough concentrations to pose an explosive hazard:

Unexploded Ordnance (UXO), as defined in 10 U.S.C. 101(e)(5)

Discarded military munitions (DMM), as defined in 10 U.S.C. 2710(e)(2)

MC (e.g., TNT, cyclotrimethylenetrinitramine) as defined in 10 U.S.C. 2710(e)(3),

military munitions. All ammunition products and components produced for or used by the armed forces for national defense and security, including ammunition products or components under the control of the DoD, the Coast Guard, the Department of Energy, and the National Guard.

Includes confined gaseous, liquid, and solid propellants, explosives, pyrotechnics, chemical and riot control agents, smokes, and incendiaries, including bulk explosives and chemical warfare agents, chemical munitions, rockets, guided and ballistic missiles, bombs, warheads, mortar rounds, artillery ammunition, small arms ammunition, grenades, mines, torpedoes, depth charges, cluster munitions and dispensers, demolition charges, and devices and components thereof.

Does not include wholly inert items, improvised explosives devices and nuclear weapons, nuclear devices, and nuclear components, but does include non-nuclear components of nuclear devices that are managed under the nuclear weapons program of the Department of Energy after all required sanitization operations under the Atomic Energy Act of 1954 (42 U.S.C. 2011 *et seq.*) have been completed. (See 10 U.S.C. 101(e) (4).)

MPPEH. Material owned or controlled by the DoD that, before determination of its explosives safety status, potentially contains explosives or munitions (e.g., munitions containers and packaging material; munitions debris remaining after munitions use, demilitarization, or

disposal; and range-related debris) or potentially contains a high enough concentration of explosives that the material presents an explosive hazard (e.g., equipment, drainage systems, holding tanks, piping, or ventilation ducts that were associated with munitions production, demilitarization, or disposal operations). Excluded from MPPEH are:

Military munitions and military munitions-related materials, including wholly inert components (e.g., fins, launch tubes, containers, packaging material), that are to be used or reused for their intended purpose and are within a DoD Component-established munitions management system.

Non-munitions-related material (e.g., horseshoes, rebar, other solid objects) and munitions debris that are solid metal fragments that do not realistically present an explosive hazard.

Other items that may present explosion hazards (e.g., gasoline cans, compressed gas cylinders) that are not munitions and are not intended for use as munitions. (See DoDI 4140.62.)

MRS. Defined in DoD 6055.09-M, Volume 8.

munitions debris. Defined in DoD 6055.09-M, Volume 8.

munitions response. Defined in DoD 6055.09-M, Volume 8.

munitions response area. Defined in DoD 6055.09-M, Volume 8.

range clearance. Defined in DoD 6055.09-M, Volume 8.

range-related debris. Defined in DoD 6055.09-M, Volume 8.

UXO. Defined in 10 U.S.C. 101(e)(5)(A) through (C).

UXOQP. Defined in DoD 6055.09-M, Volume 8.

UXOT. Defined in DoD 6055.09-M, Volume 8.

REFERENCES

American National Standards Institute, Association of Commercial Diving Educators-01, "National Standard for Divers – Commercial Diver Training – Minimum Standard," current edition
Code of Federal Regulations, Title 27, Section 555.26
Code of Federal Regulations, Title 29
Code of Federal Regulations, Title 40, Section 260.10
Code of Federal Regulations, Title 49, Section 172.704
DoD 6055.09-M, Volume 8, "DoD Ammunition and Explosives Safety Standards: Glossary," February 29, 2008, as amended
DoD Instruction 4140.62 "Material Potentially Presenting an Explosive Hazard," November 24, 2008, as amended
International Standards Organization 9001, "Quality management systems – Requirements," current edition
Technical Bulletin 700-2, Naval Sea Systems Command Instruction 8020.8C, Technical Order 11A-1-47, "Department of Defense Ammunition and Explosives Hazard Classification Procedures," July 30, 2012
United States Code, Title 10
United States Code, Title 18, Section 842
United States Code, Title 42, Section 2011 *et seq.*

TO: The Resident Officer in Charge of Construction (ROICC)
Camp Lejeune, North Carolina

Bank Name:
Attn:
Address:

Subj: FINANCIAL REFERENCE:

_____
(Mr. or Mrs. Last Name) **Signature (***Financial institution Manager)*:

The Resident Contracting Officer in Charge Construction (ROICC), NAVFAC MIDLANT, Camp Lejeune, North Carolina, is conducting a pre-award survey on

Contractor: _____,

P.O.C: _____,

Located: _____,

in conjunction with a Service Contract for Marine Corps Base Camp Lejeune, North Carolina.

As part of this survey, we need to confirm that the company has sufficient financial assets to fulfil this contract. Accordingly, please accept our apology for the short-term notice, but request the following information be provided to the contractor listed above for the price proposal submission package to Sharon E. Humphrey (Contract Specialist) and/or Regenia D. Guest (Contracting Officer) as soon as possible, but no later than the submission date of the proposal provided by the contractor.

a. Accounts(s):

Checking: Date Opened:_____ Average Balance: _____

Savings: Date Opened:_____ Average Balance: _____

Loans(s): Date Opened:_____ Average Balance: _____

b. Mortgage Cust#_____ Obligation#_____

c. Customer's Line of Credit:_____

d. Would you be willing to grant this customer a Line of Credit Increase or loan, if required?

_____YES _____NO

If no, please briefly explain:
(Please note that the account balances may be provided by indicating a range, or only the number of figures, e.g. low five figures, high four figures, etc.)

Should you have any questions concerning this request, please email Sharon E. Humphrey at Sharon.e.humphrey.civ@us.navy.mil and/or Regenia.d.guest.civ@us.navy.mil. Office hours are 7:00 AM until 3:30 PM, Monday through Friday

CERTIFICATE OF CURRENT COST OR PRICING DATA

This is to certify that, to the best of my knowledge and belief, the cost or pricing data (as defined in section 2.101 of the Federal Acquisition Regulation (FAR) and required under FAR subsection 15.403-4) submitted, either actually or by specific identification in writing, to the Contracting Officer or to the Contracting Officer's Representative in support of Solicitation N40085-27-R-0004 are accurate, complete, and current as of _____. This certification includes the cost or pricing data supporting any advance agreements and forward pricing rate agreements between the offeror and the Government that are part of the proposal.

Company's Name

Signature of Company's (President or Vice President) Representative

Date

Printed Name:

Title:

If necessary, an updated Certification will be completed to reflect when price negotiations were concluded and price agreement was reached or, if applicable, another date agreed upon between the parties; that is as close as practicable to the date of agreement on price.